

Route C Bootstrap Derivation Ledger through P7

BoYu Chen

June 7, 2026

Abstract

This note records the derivations generated in Route C stages P1–P7. It is a derivation ledger, not a claim that on-shell consistency has already derived a complete GUT. The current Route C program uses the one-family Standard Model spectrum with a right-handed neutrino as baseline, builds charge-filtered symbolic four-point pole components, checks conditional residue factorization/positivity, marks Ward-identity completion requirements, compares anomaly/chiral consistency across $SU(5)$, Pati–Salam, $Spin(10)$, E_6 , and superstring-derived completions, and classifies controlled high-energy-growth and low-energy proton-bound readiness requirements.

1 Boundary of the Ledger

Route C tests whether

$$\begin{aligned} &\text{Lorentz} + \text{unitarity} + \text{factorization} + \text{Ward identities} \\ &\quad + \text{anomaly cancellation} + \text{controlled high-energy behavior} \end{aligned} \tag{1}$$

can filter conditional GUT branches. It must not be read as

$$\text{four axioms} \implies \text{the full GUT.} \tag{2}$$

The honest statement is instead

$$\text{on-shell consistency} \implies \text{filters on allowed action-level GUT branches.} \tag{3}$$

The baseline external spectrum is one left-handed family plus a right-handed-neutrino conjugate field,

$$\mathcal{F}_{\text{SM}+\nu^c} = Q \oplus L \oplus u^c \oplus d^c \oplus \nu^c \oplus e^c. \tag{4}$$

All fields are written as left-handed Weyl fields.

2 P1: External-State Charges and Anomalies

2.1 Charge Table

The P1 charge ledger fixes the Standard Model face conventions shown in Table 1.

state	$SU(3)_C$	$SU(2)_L$	Y	$B - L$	multiplicity
Q	3	2	1/6	1/3	6
L	1	2	-1/2	-1	2
u^c	$\bar{3}$	1	-2/3	-1/3	3
d^c	$\bar{3}$	1	1/3	-1/3	3
ν^c	1	1	0	1	1
e^c	1	1	1	1	1

Table 1: P1 baseline spectrum. The fields u^c, d^c, ν^c, e^c are left-handed conjugate Weyl fields, so the listed charges are the charges of those conjugate fields.

2.2 Hypercharge Normalization

For one full family plus ν^c , the hypercharge trace is

$$\begin{aligned} \text{Tr} Y^2 &= 6 \left(\frac{1}{6} \right)^2 + 2 \left(-\frac{1}{2} \right)^2 + 3 \left(-\frac{2}{3} \right)^2 + 3 \left(\frac{1}{3} \right)^2 + 1 \cdot 0^2 + 1 \cdot 1^2 \\ &= \frac{10}{3}. \end{aligned} \quad (5)$$

With $\text{Tr}_2 T_{3L}^2 = 1/2$, the weak trace is

$$\text{Tr} T_{3L}^2 = 3 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = 2. \quad (6)$$

Therefore

$$\frac{\text{Tr} Y^2}{\text{Tr} T_{3L}^2} = \frac{5}{3}, \quad (7)$$

which is the standard GUT hypercharge normalization.

2.3 Perturbative and Global Anomaly Checks

With the conventions of Table 1, the exact rational ledger verifies

$$\mathcal{A}_{SU(3)^3} = 2A(3) + A(\bar{3}) + A(\bar{3}) = 2 - 1 - 1 = 0. \quad (8)$$

The mixed color-hypercharge anomaly is

$$\mathcal{A}_{SU(3)^2 U(1)_Y} = 2T(3)\frac{1}{6} + T(\bar{3})\left(-\frac{2}{3}\right) + T(\bar{3})\frac{1}{3} = \frac{1}{6} - \frac{1}{3} + \frac{1}{6} = 0, \quad (9)$$

where $T(3) = T(\bar{3}) = 1/2$. The weak-hypercharge anomaly is

$$\mathcal{A}_{SU(2)^2 U(1)_Y} = 3T(2)\frac{1}{6} + T(2)\left(-\frac{1}{2}\right) = \frac{1}{4} - \frac{1}{4} = 0. \quad (10)$$

The gravitational and cubic hypercharge anomalies are

$$\mathcal{A}_{\text{grav}^2 U(1)_Y} = 6\frac{1}{6} + 2\left(-\frac{1}{2}\right) + 3\left(-\frac{2}{3}\right) + 3\frac{1}{3} + 0 + 1 = 0, \quad (11)$$

$$\mathcal{A}_{U(1)^3_Y} = 6\left(\frac{1}{6}\right)^3 + 2\left(-\frac{1}{2}\right)^3 + 3\left(-\frac{2}{3}\right)^3 + 3\left(\frac{1}{3}\right)^3 + 0^3 + 1^3 = 0. \quad (12)$$

The same exact ledger verifies the analogous $B - L$ and mixed $Y - B - L$ anomalies,

$$\text{SU}(3)^2(B - L), \quad \text{SU}(2)^2(B - L), \quad \text{grav}^2(B - L), \quad (B - L)^3, \quad Y^2(B - L), \quad Y(B - L)^2, \quad (13)$$

all vanish. The Witten $\text{SU}(2)$ check passes because one family contains three Q doublets and one L doublet, hence four left-handed doublets.

3 P2: Symbolic Four-Point Pole Ansatz

The first pole ansatz is an all-incoming chiral-bilinear seed,

$$\mathcal{A}_4(12|34) = \sum_X \frac{R_s(X)}{s - M_X^2} + C_6(12|34), \quad R_s(X) = g_{12X} g_{34\bar{X}}. \quad (14)$$

P2 does not declare the mediator spectrum. It only computes which symbolic pole components are not forbidden by the P1 charges and by non-Abelian conjugacy of pair products.

The generated P2 ledger gives the numerical summary

quantity	value	
pair types	21	
pair-product components	31	
tested pair-component pairings	496	(15)
allowed symbolic pole components	18	
forbidden by Abelian charge	455	
forbidden after Abelian pass by non-Abelian conjugacy	23	

The proton-operator seeds retained at this purely symbolic level are

$$QQQL : 2 \text{ allowed pole components}, \quad u^c u^c d^c e^c : 2 \text{ allowed pole components}. \quad (16)$$

The entries

$$QQu^c e^c, \quad QLu^c d^c \quad (17)$$

have zero allowed pole components in this first all-incoming symbolic seed. These zeros are not no-go theorems for full models; they are bookkeeping results for the declared P2 ansatz and charge conventions.

4 P3: Residue Factorization and Positivity

For a declared mediator sector X , let α label all two-particle channels that can couple to X . Near a positive-norm simple pole, tree-level factorization requires

$$\text{Res}_X \mathcal{A}_{\alpha\beta} = g_{\alpha X} g_{\beta X}^*. \quad (18)$$

Thus the residue matrix is a Gram matrix,

$$R_X = g_X g_X^\dagger, \quad (19)$$

and is positive semidefinite in a positive-metric Hilbert space:

$$v^\dagger R_X v = |g_X^\dagger v|^2 \geq 0. \quad (20)$$

A wrong-sign mediator metric reverses the sign and fails the unit-coupling stress test.

The P3 machine ledger checks the P2-allowed poles and reports

quantity	value
mediator sectors	26
total vertex channels	31
P2 allowed poles checked	18
factorization checks passed	18
PSD blocks under positive metric	26
largest channel-block size	3

(21)

For a unit-coupling block with n channels, the stress spectrum is

$$\text{spec}(R_X) = \{n, 0, \dots, 0\}, \quad \text{spec}(-R_X) = \{-n, 0, \dots, 0\}, \quad (22)$$

which is the numerical pattern seen in the P3 output.

5 P4: Ward-Identity Bookkeeping

For an unbroken massless gauge boson, the target Ward check is

$$\epsilon_\mu \rightarrow p_\mu, \quad p_\mu \mathcal{M}^\mu = 0. \quad (23)$$

For a broken massive vector, the target relation is the generalized Ward/Slavnov–Taylor form

$$p_\mu \mathcal{M}^\mu(V_X) = M_X \mathcal{M}(\phi_X), \quad (24)$$

so a Goldstone/Higgs/source completion is required before a true Ward identity proof can be claimed.

The P4 ledger separates unbroken SM currents from broken/off-face currents:

quantity	value
unbroken SM current sectors	3
current transitions checked	36
massless diagonal transitions ready	6
broken/off-face transitions needing completion	30
mediator sectors inspected	26
mediator sectors needing completion if vector	26

(25)

The bookkeeping-ready unbroken sectors are

$$\text{SU}(3)_C, \quad \text{SU}(2)_L, \quad U(1)_Y. \quad (26)$$

All off-face Spin(10), Pati–Salam leptoquark, and $\text{SU}(2)_R$ charged-current transitions are flagged as needing explicit generator matrices, massive-vector data, and Goldstone/Higgs/source completion.

6 P5: Anomaly and Chiral-Consistency Candidate Ledger

P5 compares candidate branches against the P1 baseline. The structural filter used for the minimal Route C field-theory comparison is

$$\begin{aligned} &\text{simple group} + \text{one irreducible family object} + \text{contains } \nu^c \\ &+ \text{no chiral SM exotics} + \text{minimal standard realization.} \end{aligned} \quad (27)$$

6.1 SU(5)

One family plus ν^c is realized as

$$10 \oplus \bar{5} \oplus 1. \quad (28)$$

The cubic anomaly cancels:

$$A(10) + A(\bar{5}) + A(1) = 1 - 1 + 0 = 0. \quad (29)$$

Also $\pi_4(\text{SU}(5)) = 0$ in the standard four-dimensional global-anomaly check. However, the family is not one irreducible object; therefore SU(5) is anomaly-consistent but fails the single-object filter.

6.2 Pati–Salam

The standard Pati–Salam family is

$$(4, 2, 1) \oplus (\bar{4}, 1, 2) \quad \text{under} \quad \text{SU}(4)_C \times \text{SU}(2)_L \times \text{SU}(2)_R. \quad (30)$$

The $\text{SU}(4)_C^3$ anomaly cancels because

$$2A(4) + 2A(\bar{4}) = 2 - 2 = 0. \quad (31)$$

The Witten checks pass because $\dim(4) = 4$ gives four $\text{SU}(2)_L$ doublets and four $\text{SU}(2)_R$ doublets. Pati–Salam is physically natural and anomaly-consistent, but it is a product group and not a single irreducible family object under a simple group.

6.3 Spin(10)

The standard Spin(10) family is the half-spinor

$$16. \quad (32)$$

It contains the full

$$Q \oplus L \oplus u^c \oplus d^c \oplus \nu^c \oplus e^c \quad (33)$$

after restriction to the Standard Model face. The D_5 algebra has no independent cubic gauge-anomaly invariant in this standard check, and $\pi_4(\text{Spin}(10)) = 0$. Therefore Spin(10) : 16 is anomaly-safe and is the standard minimal single-object survivor under the stated Route C pre-filter.

6.4 E6

The E_6 fundamental representation restricts as

$$27 \longrightarrow 16 \oplus 10 \oplus 1 \quad \text{under Spin}(10). \quad (34)$$

It contains a Spin(10) family, but also contains extra $10 \oplus 1$ states. Since E_6 has no independent cubic gauge-anomaly invariant for the standard 27, the anomaly status is acceptable, but the branch requires an explicit lifting/vectorlike audit for the extra states. It is therefore not the minimal Route C survivor.

6.5 Superstring-Derived Completions

Superstring-derived completions are not a single four-dimensional field-theory group. They form a UV-completion class. A string-derived branch must supply compactification-specific data:

$$\text{Green-Schwarz/tadpole/modular consistency,} \quad \text{massless spectrum,} \quad \text{chiral index,} \quad (35)$$

together with exotics lifting and matching to the Route-A representation/index skeleton. Such branches may provide stronger high-energy softness or tower structure, but P5 does not treat them as a single fixed four-dimensional candidate.

6.6 P5 Numerical Summary

The generated P5 ledger reports

quantity	value
candidates inspected	5
field-theory candidates	4
UV-completion classes	1
P1 baseline anomalies pass	true
minimal single-object survivor	Spin(10)

(36)

This result should be read narrowly: passing anomaly checks alone does not select a GUT. The selection of Spin(10) occurs only after imposing the Route C minimal single-object family filter.

7 P6: Controlled High-Energy-Growth Audit

P6 consumes the P3 residue blocks and the P4 Ward-completion ledger. Its purpose is not to compute a full four-point amplitude. Its purpose is to classify which symbolic branches would have uncontrolled high-energy growth if treated as isolated finite broken-vector exchange.

For a finite massive vector, the dangerous contribution is the longitudinal propagator term. A conservative schematic form is

$$\mathcal{A}_L \sim \frac{(J \cdot p)(J' \cdot p)}{M_X^2(s - M_X^2)}. \quad (37)$$

If the broken current is not completed by the corresponding Goldstone/Higgs sector, this can scale as

$$\mathcal{A}_L \sim \frac{s}{M_X^2} \quad \text{or worse} \quad (38)$$

in the high-energy regime. Therefore the required broken-vector target is again the generalized Ward/Slavnov-Taylor identity

$$p_\mu \mathcal{M}^\mu(V_X) = M_X \mathcal{M}(\phi_X). \quad (39)$$

The generated P6 ledger reports

quantity	value
mediator sectors inspected	26
finite-vector branches needing completion	26
unbroken-vector branches bookkeeping-ready	0
scalar/auxiliary branches without longitudinal-vector risk	26
contact-only components needing UV completion	18
broken transitions needing completion	30
tower-like UV forced now	false

(40)

Thus every current P3 mediator sector, if interpreted as a finite broken vector, needs explicit finite-completion data:

$$\{\text{broken generator, } M_X, \phi_X, \text{Higgs/source sector, symmetry-fixed contact/numerator terms}\}. \quad (41)$$

If a symbolic mediator is instead interpreted as scalar or auxiliary exchange, the longitudinal-vector Ward problem is absent. This is only a bookkeeping statement; spin-statistics, action existence, low-energy matching, and perturbative unitarity remain separate checks.

A contact-only four-fermion EFT normalization generically scales like

$$\mathcal{A}_{\text{contact}} \sim \frac{s}{\Lambda^2}, \quad (42)$$

so contact terms may encode low-energy matching but cannot by themselves serve as high-energy softness proofs.

Finally, P6 does not force a string or tower completion. It states the conditional comparison rule:

$$\begin{array}{l} \text{finite Goldstone/Higgs/source completion fails} \\ \text{tower-like UV completion becomes a serious comparison class.} \end{array} \implies \quad (43)$$

Superstring-derived completions therefore remain comparison branches, not P6 conclusions.

8 P7: Low-Energy Matching and Proton-Bound Readiness

P7 converts the P2/P3 symbolic pole data into conditional low-energy dimension-six matching rows. It respects the P6 gate: a vector interpretation of a broken mediator cannot be promoted to a physical Wilson coefficient until the finite Goldstone/Higgs/source completion has been supplied.

For a symbolic pole with mediator X , P7 records the matching template

$$C_6[\mathcal{O}; X] = \frac{g_{\text{left},X} g_{\text{right},X}^*}{M_X^2}. \quad (44)$$

This is only a gauge-basis symbolic coefficient. It is not yet a physical proton-decay Wilson coefficient.

Using the left-handed Weyl charge convention, the global bookkeeping assigns

$$B(Q) = \frac{1}{3}, \quad B(u^c) = B(d^c) = -\frac{1}{3}, \quad L(L) = 1, \quad L(\nu^c) = L(e^c) = -1. \quad (45)$$

The generated P7 ledger reports

quantity	value
allowed symbolic poles processed	18
B and L conserving poles	12
standard BNV seed poles	4
BNV-with-sterile-neutrino poles	2
baryon-violating symbolic poles	6
BNV vector poles blocked by P6 completion	6
physical proton bounds evaluable now	0
all physical bounds conditional	true

(46)

The standard baryon-violating seeds include

$$QQQL, \quad u^c u^c d^c e^c, \quad (47)$$

and the sterile-neutrino branch includes

$$u^c d^c d^c \nu^c. \quad (48)$$

All of these preserve $B - L$ in the P1 bookkeeping.

The physical proton-bound interface is recorded as

$$\Gamma(p \rightarrow a) = \sum_{i,j} C_i^{\text{phys}} C_j^{\text{phys}*} H_{ij}^{(a)}, \quad (49)$$

where $H_{ij}^{(a)}$ contains phase space, RG evolution, chiral reduction, and hadronic matrix elements for channel a . A lifetime lower bound implies

$$(C^{\text{phys}})^\dagger H^{(a)} C^{\text{phys}} < \frac{1}{\tau_a^{\text{exp}}}. \quad (50)$$

For a single coefficient this reduces schematically to

$$|C_i^{\text{phys}}| < \sqrt{\frac{1}{\tau_a^{\text{exp}} H_{ii}^{(a)}}}. \quad (51)$$

P7 deliberately does not insert numerical proton limits. Numerical lifetimes require:

P6-completed high-energy branch, mediator masses and couplings, physical flavor rotations,
operator basis and chiral contractions, RG factors, lattice/chiral matrix elements, τ_a^{exp} .
(52)

Thus P7 is a proton-bound readiness report, not a proton-lifetime calculation.

9 Current Status and Next Check

The P1–P7 chain currently establishes:

1. the exact P1 charge/anomaly baseline for one $SM + \nu^c$ family;
2. the P2 charge-filtered symbolic pole components;
3. the P3 conditional residue-factorization and PSD statement;
4. the P4 Ward-identity completion ledger for broken/off-face sectors;
5. the P5 anomaly/chiral-consistency comparison across candidate branches.
6. the P6 high-energy-growth ledger for finite-vector, scalar/auxiliary, contact-only, and tower-like comparison branches.
7. the P7 conditional low-energy matching and proton-bound readiness ledger.

The next natural stage is P8: a candidate-theory comparison scorecard. It should compare $SU(5)$, Pati–Salam, $Spin(10)$, E_6 , and superstring-derived completions using the explicit P1–P7 gates, rather than using anomaly cancellation or minimality alone.